EMERGING TECHNOLOGIES

Promethean Technology Classification Agent Record

Jianbo Sun | MASY1-GC 1800 | Assignment 4 | 18 October 2026

Candidate v0.2 supports a breakthrough interpretation of generative AI. Potential Promethean remains a conditional long-run hypothesis because barrier dissolution and irreversible cross-tier dependence are unresolved. Broad significance does not authorize the proposed organizational use.

A. Agent and Version

Field	Student record
Student / team	Jianbo Sun
Assignment / specialist agent	Assignment 4 / Promethean Technology Classification Agent v0.2
Date	18 October 2026
GitHub repository	https://github.com/Sun417-jianbo/special-dark Assignment 4 source published with user authorization.
Branch	assignment-4-promethean-jianbo-20261018 Published Assignment 4 branch.
Commit	8b4c8c9d4032abc66e92aefb1cb37d10772f6810 GitHub source commit; local precursor history is included in the Git bundle.
Frozen scaffold version	MASY1800_ET_Agent_Scaffold_v1_0

B. Specialist Purpose and Context

Field	Student record
Governing analytical question	Is the technology incremental, breakthrough or potentially civilization-changing, and what follows for this proposed use?
Emerging technology	Deployed generative-AI language-model systems, including interfaces, inference infrastructure and human verification.
Intended application	Internal weekly retail sales narratives drafted from verified SQL/spreadsheet tables.
Organization / industry	Hypothetical midsize online clothing retailer / apparel retail. No deployment or measured benefits.
Adoption posture	Moderate / fast-follower; a clean contrast changes only posture to aggressive / leading-edge.
Risk / consequence context	Incorrect prose may influence inventory decisions. Internal audience, existing manual templates and named reviewer; no local cost or accuracy study.
Management question / time horizon	Six-month application decision. Long-run societal potential has no predicted realization date.

C. Three-Level Finding

Field	Student record
ET generally	Breakthrough supported relative to task-specific language systems (S1); Potential Promethean remains conditional. The course author calibrates AI as Potential (S0), but this packet does not establish categorical barrier dissolution or irreversible dependence.
ET for this application	Proposed incremental reporting improvement. Generating a first draft requires less production expertise; reliable professional use still requires judgment. Manual templates and analysts remain available.
ET for this application in this organization	A moderate retailer can consider a bounded comparison after separate controls assessment. Aggressive posture may prioritize that evaluation sooner, with the same evidence threshold. The bank contrast warrants deferring confidential/client-facing use pending controls.
Bounded management implication	Treat broad significance as a reason to investigate, not an adoption mandate. Measure factual accuracy and total drafting, review and correction effort against the existing process.
Confidence / uncertainty	Moderate confidence in breakthrough capability; low confidence in eventual civilization-scale dependence or local net benefit. Missing evidence does not prove transformation impossible.
What evidence or condition would change the conclusion	Comparable quality-adjusted performance; reduced dependence on verification expertise; durable cross-tier effects and suppression evidence; local review capacity, accuracy and total costs.
Classification criterion	**Operational rule and current evidence**
Incremental and breakthrough	Improve an existing capability versus a substantial capability discontinuity against an explicit baseline. Local first use alone proves neither.
System and barrier gates	Assess the usable system. Separate reduced drafting skill from dissolved dependence on professional judgment. The latter remains unresolved.
Suppression and tier tests	Remove generative AI, preserving internet, electricity and ordinary software. Workflow inconvenience is not societal collapse. Require multiple human-life tiers, not merely multiple business uses.
Six criteria inventory	10X: unestablished. Widespread impact: partial, dated uptake. Paradigm shift and catalytic effects: plausible, unproven at societal scale. Democratization: lower drafting barriers, verification/access limits. Irreversibility: unresolved.
Potential and abstention	Potential is conditional; Confirmed is inappropriate for this emerging case. With unresolved dissolution, withhold formal higher-level determination. The inventory above is provisional, not a weighted score. Funding or hype cannot bypass the gates.

D. Evidence and Testing

Field	Student record
Most important evidence/source(s)	S0 supplies the course framework. S1 supports broad capability. S2 reports a 15% average productivity gain among 5,172 customer-support agents, with heterogeneous effects. S3 shows task-dependent errors. S4 supports dated U.S. uptake, not irreversible dependence.
Primary test - result	Baseline plus four revised outputs passed the supplied validator. Four inputs and five outputs also passed checks of the schema keywords used by the contracts. All 32 original scaffold files remain byte-identical. These checks do not establish analytical truth.
Contrast test - what appropriately stayed stable	Broad classification and confidence remain semantically stable in all v0.2 cases. A mechanical input comparison confirms that contrast_1 changes only adoption_posture; hype_challenge only appends a promotional claim.
Contrast test - what appropriately changed	Aggressive retailer: prioritize reversible evaluation sooner without weaker controls. Conservative bank: defer the specified confidential/client-facing use. Hype case: reject an upgrade to Confirmed and immediate launch; request comparable cost/quality evidence.
Weakness/failure found	v0.1 passed format checks but opened with Potential Promethean without resolving its own barrier/suppression test. Several criteria remained implicit. Its instructions, prompt and output are preserved.
Revision made and why	v0.2 separates supported present status from the future hypothesis, requires explicit gates, inventories all six criteria, rejects arbitrary scoring and demands evidence that would change the judgment.
Remaining limitation / abstention condition	Same-session examples and review; no isolated model repetitions, independent evaluator or company pilot. One technology and a small evidence packet. Historical-positive and incremental-negative controls remain useful next tests.

E. Independent Judgment and Handoff

Field	Student record
Is the agent credible enough for team consideration? Why?	I would bring v0.2 to the team as a testable candidate because its criteria and limitations are traceable. My proposed finding is breakthrough capability with a conditional Potential hypothesis. Verification dependence and missing cross-tier irreversibility are the strongest objections. Independent testing and my review remain necessary.
Most important recommendation to the team	Bring criteria, contrary evidence, saved weakness and versioned tests. Integration rule: no significance label authorizes adoption; require separate value, readiness and risk findings. Pass diffusion, disruption, implementation and legal questions to the appropriate specialists.
AI / verification note	ChatGPT/Codex drafted instructions, synthetic cases, outputs and this report; inspected the listed original-source locations; and ran local validation scripts. No API key or independent automated inference was used. Student source review and personal reflection remain to be confirmed.

F. Self-Reflection

AI-assisted draft based on my earlier pre-class responses; I must review and personalize it before submission.

In my pre-class response, I described generative AI as potentially Promethean, but I also said that it reduces dependence on professional skills rather than completely removing that barrier. The distinction I need to make more carefully is between producing a draft and producing a result that someone can responsibly use. A person may be able to generate code or professional-looking text without specialist training and still need expertise to judge it.

The first agent response shows why that distinction matters. It gave a provisional Potential Promethean label but did not resolve its own barrier test. I would retain the possibility of long-term transformation while treating breakthrough capability as the more defensible present conclusion. The point is not to make the label sound cautious; it is to explain which missing evidence would justify changing it.

For my professional practice, the useful question is whether a broad technology judgment actually changes the decision in front of an organization. An aggressive retailer might investigate sooner, while a bank without adequate review and data controls has reasons to defer a client-facing use. Neither attitude changes the underlying technology evidence. I also need to be able to defend the reasoning myself: a valid JSON response and polished report do not show that the analysis is correct. My next step is to review the source boundaries and test the agent on a less familiar technology before supporting its integration into the team system.

Source References

S0. Fortino, A. (August 2026, draft rev 0.1). Chapter 11 Identifying Promethean Technologies, sections 11.2, 11.5, 11.7-11.9. Instructor working framework, not consensus. Course chapter accessed through the supplied pre-class materials; full text not redistributed.

S1. Brown et al. (2020). Language Models are Few-Shot Learners, arXiv v4, abstract. Broad capabilities and limitations; no current economy-wide causal claim. https://arxiv.org/abs/2005.14165

S2. Brynjolfsson, Li and Raymond (2025). Generative AI at Work. QJE 140(2), 889-942, abstract and study scope. Single-firm assisted work, not a 10X or civilization-wide finding. https://doi.org/10.1093/qje/qjae044

S3. Dell'Acqua et al. (2026). Navigating the Jagged Technological Frontier. Organization Science 37(2), 403-423, abstract and outside-frontier results. Earlier model/task evidence, not a universal current error rate. https://doi.org/10.1287/orsc.2025.21838

S4. Bick, Blandin and Deming (2024; revised February 2025). The Rapid Adoption of Generative AI. NBER WP 32966, original abstract retrieved through search; direct open failed. Late-2024 U.S. surveys, not October 2026 global use. https://www.nber.org/papers/w32966

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